

SML 2026, Winter, Quiz 2, Dur. 30 mins. Total marks
10

Name:

RollNo.

Instructions

Q1 is one mark.

Q2 to Q7 are 1.5 marks each.

Q8 is bonus question

NEGATIVE marking. For wrong answer, half marks will be deducted.

Mark only the correct answer. No justification is needed.

Q1. In 2-class Gaussian discriminant analysis with unequal covariance, the decision boundary is:

(A) Always spherical

(B) Linear

(C) Piecewise linear

☒ (D) Quadratic

Q2. Consider a binary classification setup with multivariate Gaussian likelihoods. Increasing prior for class 1 will result in:

(A) No change

(B) Shift decision boundary toward mean of class 1

☒ (C) Shift decision boundary toward mean of class 2

(D) Rotates decision boundary only

Q3. In case of binary classification with multivariate Bernoulli likelihood, j -th dimension's probabilities are same. Which of the following is correct?

☒ (A) Zero contribution towards decision boundary

(B) Decision boundary will become non-linear

(C) Removing j -th feature changes the decision.

(D) None of the above

Q4. Uniform prior over the domain of the parameter to be estimated implies:

(A) Unbiased estimate

(B) Independent of the number of given samples

☒ (C) MAP=MLE

(D) None of the above

Q5. Consider Bernoulli observations with $n = 20$ trials and $s = 7$ successes, where a success means that observed value is 1. Assume a Beta prior $\text{Beta}(\alpha, \beta)$ with $\alpha = 3$, $\beta = 5$. The MAP estimate of θ is _____.

(A) $\frac{7}{20}$

(B) $\frac{10}{28}$

(C) $\frac{8}{22}$

☒ (D) $\frac{9}{26}$

Q6. Bhattacharyya error bound corresponds to Chernoff error bound at:

(A) $s = 0$

(B) $s = 1$

☒ (C) $s = \frac{1}{2}$

(D) $s \rightarrow \infty$

Q7. Consider binary classification with $y \in \{0, 1\}$ and exponential class-conditionals:

$$p(x | y = k) = \theta_k e^{-\theta_k x}, \quad x \geq 0, \quad k \in \{0, 1\}.$$

Let $\theta_0 = 0.5$, $\theta_1 = 2$, and priors $\pi_0 = 0.7$, $\pi_1 = 0.3$. The Bayes decision boundary can be written as

$$wx + b = 0,$$

where w and b are constants. Choose the correct pair (w, b) .

☒ (A) $w = \theta_0 - \theta_1, \quad b = \log \frac{\pi_1 \theta_1}{\pi_0 \theta_0}$

(B) $w = \theta_1 - \theta_0, \quad b = \log \frac{\pi_1 \theta_1}{\pi_0 \theta_0}$

(C) $w = \theta_0 - \theta_1, \quad b = \log \frac{\pi_0 \theta_0}{\pi_1 \theta_1}$

(D) None of the above

Q8. [Bonus] Let $x_1, \dots, x_n \stackrel{ind}{\sim} \mathcal{N}(\mu, 1)$ with unknown μ . Assume $\mu \in \{0, 1\}$ and the prior on μ is Bernoulli pmf. Let $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$. The equivalent MAP maximization problem can be written in which of the following forms?

(A)

$$\mu_{\text{MAP}} = \arg \max_{\mu \in \{0,1\}} \left[-\frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2 + \log \theta + -\mu \theta \right].$$

(B)

$$\mu_{\text{MAP}} = \arg \max_{\mu \in \{0,1\}} \left[\frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2 + \mu \log \theta + (1 - \mu) \log(1 - \theta) \right].$$

(C)

$$\mu_{\text{MAP}} = \arg \max_{\mu \in \{0,1\}} \left[\mu \log \frac{\theta}{1 - \theta} + \sum_{i=1}^n \left(x_i^2 - \frac{1}{2} \right) \mu^2 \right].$$

~~(D)~~ None of the above